National University of Modern Languages, Islamabad

Speech Processing Lab Report

Lab 8: Cepstrum, Liftering & Excitation Reconstruction

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Objective

The objective of this lab is to:

- Read two speech signals and ensure matching sampling rates.
- Compute the real cepstrum using FFT and IFFT.
- Apply liftering to separate vocal-tract envelope and excitation.
- Reconstruct deconvolved excitation signals.
- Visualize the cepstrum and excitation signals.

MATLAB Code: Cepstrum, Liftering & Excitation Reconstruction

```
clc; clear; close all;

%% Step 1: Read speech signals
disp('Reading speech files...');

[x1, fs1] = audioread('D:\matlab-jd\speech1.wav');
[x2, fs2] = audioread('D:\matlab-jd\speech2.wav');

% Match sampling rates
if fs1 ~= fs2
    disp('Sampling frequencies do not match | resampling second signal...');
    x2 = resample(x2, fs1, fs2);
    fs2 = fs1;
end

%% Function for cepstrum computation
computeCepstrum = @(x,fs) struct( ...
    'x', x(:), ... % force column vector
    'fs', fs, ...
    'X', fft(x(:)), ...
    'logX', log(abs(fft(x(:))) + eps), ...
    'cep', real(ifft(log(abs(fft(x(:))) + eps))) ...
);

%% Step 2: Compute cepstrum
```

```
disp('Computing cepstrum...');

c1 = computeCepstrum(x1, fs1);
c2 = computeCepstrum(x2, fs2);

%% Step 3: Plot time domain and cepstrum
figure;
subplot(2,1,1); plot(c1.x); title('speech1.wav { Time Domain');
subplot(2,1,2); plot(c1.cep); title('speech1.wav { Cepstrum');

figure;
subplot(2,1,1); plot(c2.x); title('speech2.wav { Time Domain');
subplot(2,1,2); plot(c2.cep); title('speech2.wav { Cepstrum');

%% Step 4: Liftering (High-time window)
L = 30;

% Create lifter of correct length for each file
hl1 = [zeros(L,1); ones(length(c1.cep)-2*L,1); zeros(L,1)];
hl2 = [zeros(L,1); ones(length(c2.cep)-2*L,1); zeros(L,1)];

% Ensure lengths match EXACTLY
hl1 = hl1(1:length(c1.cep));
hl2 = hl2(1:length(c2.cep));

cep1_exc = c1.cep .* hl1;
cep2_exc = c2.cep .* hl2;

%% Step 5: Reconstruct excitation signals
X1_exc = exp(fft(cep1_exc));
X2_exc = exp(fft(cep2_exc));

x1_exc = real(ifft(X1_exc));
x2_exc = real(ifft(X2_exc));

%% Step 6: Plot deconvolved signals
figure;
subplot(2,1,1); plot(x1_exc); title('Deconvolved Excitation { speech1.wav');
subplot(2,1,2); plot(x2_exc); title('Deconvolved Excitation { speech2.wav');
```

```
disp('Processing complete.');
```

Explanation

- The two speech files are loaded and sampling rates are equalized.
- Cepstrum is computed using:

$$c[n] = Re\left\{IFFT\left(\log |X[k]|\right)\right\}$$

- Cepstrum shows vocal-tract envelope (low quefrencies) and excitation (high quefrencies).
- Liftering isolates the excitation by zeroing low-time peaks.
- Exponential + FFT inversion reconstructs excitation:

$$x_{exc} = IFFT\left(e^{FFT(cep_{exc})}\right)$$

- Excitation signals are plotted to observe pitch-related impulses.

Output Images

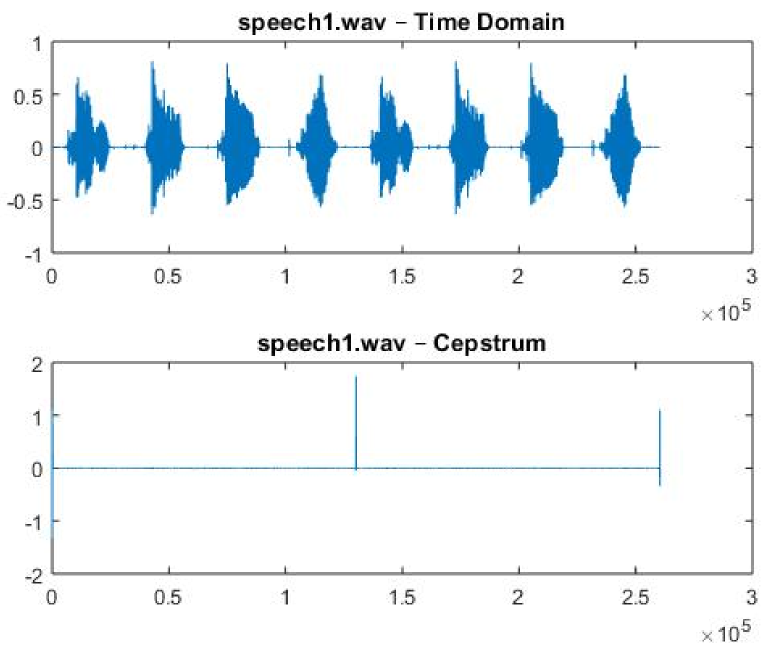

Figure 1: Time Domain and Cepstrum of speech1.wav

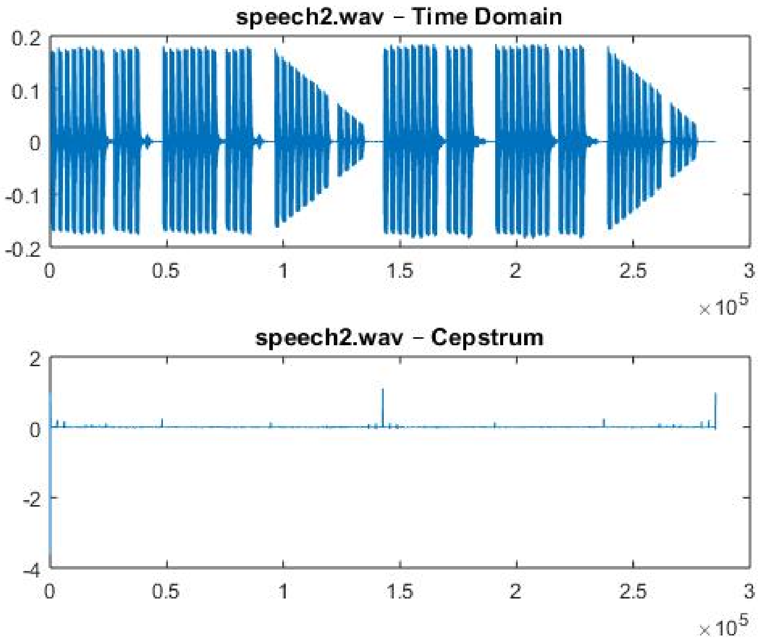

Figure 2: Time Domain and Cepstrum of speech2.wav

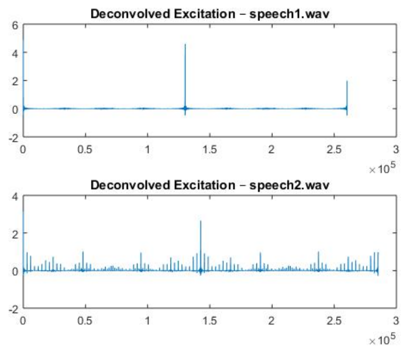

Figure 3: Reconstructed Excitation Signals After Liftering

Conclusion

This lab demonstrated cepstrum computation, liftering, and excitation signal recovery. By removing the low-time components of the cepstrum, we isolated the source excitation. The reconstructed excitation signals clearly show pitch impulses, validating the cepstral deconvolution method. This technique is widely used in speech coding, pitch detection, and speech analysis.